Executive Decision Report

Retail Performance & Customer Experiment Analysis

RABTECH Academy · Data Analytics & Business Intelligence Internship

Decision focus: Evaluate customer conversion, order value, discount strategy, and customer experience using the retail A/B-test-style dataset.

Prepared as a capstone template. Populate the findings and ROI estimates with verified outputs from Tasks 03–05 before final submission.

1. Executive Summary

• Use the dashboard and EDA notebook to compare conversion, revenue, average order value, returns, and satisfaction across experiment groups.

• Do not roll out a discount based only on conversion lift; check contribution margin, return rate, and customer experience.

• Recommended decision: run a controlled pilot and expand only if incremental profit is positive and guardrails remain healthy.

2. Business Question & Decision Needed

Should the business expand a discount offer, keep the current control strategy, or test a more targeted promotion? The decision should balance incremental conversions against discount cost, fulfilment cost, and potential returns.

- Primary outcome: conversion rate.
- Secondary outcomes: average order value, revenue per visitor, return rate, satisfaction.
- Decision owner: commercial leadership / growth team.

3. Dataset & Scope

- Source: retail A/B-test-style dataset downloaded from Kaggle and cleaned for Task 03.
- Unit of analysis: confirm whether each row represents a customer, session, or experiment assignment before interpreting results.
- Segments: experiment group, region, device, traffic source, loyalty tier.
- Scope limitation: results apply to the supplied dataset; do not generalize to a real business without validating source and sampling.

4. Methodology

- Task 03: inspect missing values, duplicates, data types, and possible outliers; preserve raw source separately.
- Task 04: descriptive statistics, histograms, box plots, correlation analysis, and three hypothesis tests.
- Task 05: interactive KPI dashboard with filters for experiment group, region, and device.
- Use a 5% significance threshold as a convention; report effect sizes and sample sizes alongside p-values.

5. Executive KPI Scorecard

KPI	Definition / calculation
Revenue	Sum of order value for confirmed purchases; validate whether order_value is zero for non-purchases.
Conversion rate	Purchases or conversion flags divided by eligible observations.
Average order value	Revenue divided by number of orders, not total visitors.
Return rate	Returned orders divided by orders; define denominator clearly.
Customer acquisition cost (CAC)	Acquisition spend divided by new customers acquired; unavailable unless spend data exists.

6. Revenue & Conversion Drivers

- Compare conversion rate by experiment group and week.
- Review performance by traffic source, region, and device to locate promising segments.
- Distinguish increased conversion from increased profit: a discount can increase orders while reducing margin.
- Add verified figures from Task 04/05 outputs before submission.

7. Discount Experiment Assessment

- Compare Control with Discount10 using a two-proportion test for conversion.
- Compare order value among purchasers using Welch's t-test, while noting selection effects among purchasers.
- Review satisfaction and returns as guardrail metrics.
- Report absolute percentage-point lift and relative lift separately; include confidence intervals where possible.

8. Risk Areas & Controls

Risk	Mitigation
Discount erodes margin	Estimate gross margin and discount cost before rollout.
Segment imbalance	Check sample sizes and baseline comparability by group.
Misleading correlation	Treat correlations as associations, not causal effects.
Data quality issues	Document exclusions, missing values, duplicates, and transformations.
Returns / satisfaction decline	Set pre-agreed guardrails and stop rules for the pilot.

9. Strategic Recommendation 1 — Controlled Pilot

- Continue a randomized, time-bounded pilot rather than broad rollout.
- Predefine primary metric, guardrails, sample size, and decision date.
- Expand only if the incremental profit estimate is positive and key guardrails are met.

10. Strategic Recommendation 2 — Targeted Offers

- Use segment-level results as hypotheses for targeted follow-up tests, not as proof that a segment will respond.
- Compare eligible customer groups on conversion, order value, returns, and satisfaction.
- Avoid targeting based on sensitive or inappropriate attributes.

11. Strategic Recommendation 3 — Measurement & Dashboard

- Maintain a single KPI definition sheet and consistent denominators.
- Review weekly trends and breakdowns by region, device, traffic source, and experiment group.
- Automate refreshes only after data definitions and quality checks are stable.

12. ROI Projection Framework

Input	Formula / requirement
Incremental orders	Eligible visitors × estimated absolute conversion lift
Incremental revenue	Incremental orders × expected average order value
Incremental contribution	Incremental revenue × contribution margin rate
Promotion cost	Discount cost on all eligible orders + campaign costs
Net incremental profit	Incremental contribution – promotion cost – incremental fulfilment/return costs
ROI	Net incremental profit ÷ total incremental investment

13. Implementation Timeline

	Deliverable
	Confirm dataset grain, KPI definitions, and quality checks
	Pre-register hypothesis, sample size, guardrails
4	Run controlled experiment and monitor quality
	Calculate effect sizes, confidence intervals, ROI
	Scale, iterate, or stop based on agreed criteria

14. Decision Framework

- Scale: statistically and commercially meaningful lift, positive incremental profit, and guardrails within limits.
- Iterate: promising but inconclusive results; improve test design or target a narrower hypothesis.
- Stop: no meaningful benefit, negative ROI, or deterioration in returns/satisfaction.
- Final decision should be based on verified Task 04/05 results, not assumptions in this template.

15. Next Steps & Appendix

- Insert actual conversion rates, order values, sample sizes, p-values, and dashboard screenshots.
- Replace the ROI framework with a numeric scenario only when cost and margin inputs are available.
- Attach the final analytical notebook and deployed dashboard link as requested.
- Record assumptions, data source, date, and limitations.

Submission note

This report intentionally avoids inventing numerical findings or ROI values. Before submitting, replace placeholders and generic recommendations with the verified results from your analysis.